Singapore School Kinderland

Secondary 1 · Computer Science · Term Plan · Term 2

	Ch 2.2: Methods of Error Detection (Pg 54-62)				Ch 2.3: Encryption & Security (Pg 63-74)				Ch 2.2 Review & Practice		Ch 2.3 Review & Practice		Term Review		Exam
	Wk 1		Wk 2		Wk 3		Wk 4		Wk 5		Wk 6		Wk 7		Wk 8
	Lesson Objective	Content	Lesson Objective	Content	Lesson Objective	Content	Lesson Objective	Content	Lesson Objective	Content	Lesson Objective	Content	Lesson Objective	Content	
Day 1	St will explain reasons for data transmission errors. St will define Even and Odd Parity and calculate parity bits for 8-bit bytes.	TB Pg 54-55 2.2.1 Parity Bit Checks	St will explain Checksum algorithms: sender block sum calculation, transmission, and recipient re-calculation & comparison.	TB Pg 58-59 2.2.3 Checksum Algorithm WS Checksum Tasks	St will explain Check Digits (detecting entry errors: transposition, phonetic) and calculate ISBN-13 and Modulo 11 check digits.	TB Pg 60-61 2.2.5 Check Digits Modulo 11 WS ISBN Math	St will define Encryption, Plain text, Cipher text, and explain Symmetric Encryption using a single shared secret key.	TB Pg 63-65 2.3.1 Symmetric Encryption Caesar/OTP	St will explain Asymmetric Encryption: public and private key pairs, encryption using public key and decryption with private key.	TB Pg 68-70 2.3.3 Public/Private Key Encryption	St will solve Cambridge past paper questions on error detection (parity byte correction, checksum vs parity, ARQ flow scenarios).	WS Pg 1-3 Ch 2.2 Cambridge Past Paper Questions	St will revise all error detection methods and encryption concepts through structured group Q&A and timed topical quiz.	WS Comprehensive Error Detection & Encryption Review	Term Exam
Day 2	St will analyze Parity Blocks (horizontal & vertical parity bytes) and pinpoint corrupted bit locations using parity coordinate grids.	TB Pg 56-57 2.2.2 Parity Blocks Lab WS Error Matrix	St will define Echo Check (sending data back for comparison) and analyze its limitation (error during echoing vs transmission).	TB Pg 59-60 2.2.4 Echo Checks & Limitations	St will evaluate Automatic Repeat Requests (ARQ): acknowledgements (ACK), negative acknowledgements (NAK), and timeout limits.	TB Pg 61-62 2.2.6 ARQ Protocol WS Protocol Flowchart	St will evaluate security vulnerabilities of symmetric encryption: key distribution problem, interception, and brute-force attacks.	TB Pg 66-67 2.3.2 Key Distribution WS Symmetric Tasks	St will explain the purpose of Hashing, Digital Signatures, and Digital Certificates (SSL/TLS) for authenticating sender identity.	TB Pg 71-74 2.3.4 Digital Signatures & Certificates WS SSL	St will solve past paper questions on encryption (symmetric vs asymmetric comparison table, public/private key scenarios).	WS Pg 4-6 Ch 2.3 Cambridge Past Paper Questions	St will complete full mock practice worksheet covering all question formats (MCQs, structured explanations, error-finding tables).	WS Term 2 Mock Practice Assessment	Results & Feedback